

507. Perfect Number

Input: num = 28

Output: true

Explanation: $28 = 1 + 2 + 4 + 7 + 14$
1, 2, 4, 7, and 14 are all divisors of 28.

```
public boolean checkPerfectNumber(int n) {  
    int sum = 0;  
    for(int i=1; i<=n/i; i++){  
        if(n%i==0){  
            if(i==n/i){  
                sum += i;  
            }else{  
                sum += i+n/i;  
            }  
        }  
    }  
    sum = sum-n;  
    return n==sum;  
}
```

H/W Prime Number.

ip: 13

op: True.

```
public static boolean PrimeNumber(int n){  
    int cnt = 0;  
    for(int i=1; i<=n/i; i++){  
        if(n%i==0){  
            if(i==n/i) cnt++;  
            else cnt+=2;  
            if(cnt>2) return false;  
        }  
    }  
    return true;  
}
```

Array

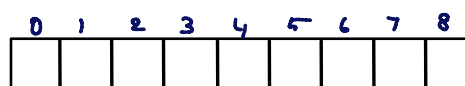
Collection of elements with same datatype.

Array has continuous memory allocation.

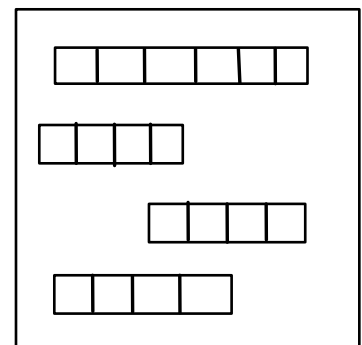
Can access array using index.

Index starts from zero.

Array is a fixed size.



0 to len-1.



Syntax:-

manually : Datatype ArrayName [] = { 10, 20, 30, 40, 50 };

It helps to store
collection of element.

user Input:- Datatype ArrayName[] = new Datatype [Size];

- 1) Create a size for an array.
- 2) Create an array and pass the size.

array length $\Rightarrow$ ArrayName.length.

arr[i] $\rightarrow$ value in that index.

i $\rightarrow$ It represent the index.

```
import java.util.*;
public class Main{
    public static void main(String[] args) {
        int arr[] = {10, 20, 30, 40};
        for(int i=0; i<arr.length; i++){//i-> index
            System.out.print(arr[i]+" ");
        }
        System.out.println();
        //reverse
        for(int i=arr.length-1; i>=0; i--){//i-> index
            System.out.print(arr[i]+" ");
        }
    }
}
```

Declaring Array for user input:-

```
import java.util.*;
public class Main{
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int n = sc.nextInt();
        int arr[] = new int[n];  $\rightarrow$ 

|   |   |   |   |
|---|---|---|---|
| 0 | 0 | 0 | 0 |
|---|---|---|---|

 Initial val of an array.
        //To get the values from the user
        for(int i=0; i<n; i++){// 1 2 3 4 5
            arr[i] = sc.nextInt();  $\rightarrow$ 

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
|---|---|---|---|

 To store the val.
        }
        //To print the values from the user
        for(int i=0; i<n; i++){
            System.out.print(arr[i]+" ");  $\rightarrow$ 

|   |   |   |   |
|---|---|---|---|
| 1 | 2 | 3 | 4 |
|---|---|---|---|

 To print the val.
        }
    }
}
```

Array updation using function: (without return type)

```
import java.util.*;
public class Main {
    public static void UpdateArray(int a[]) {
        for(int i=0; i<a.length; i++) {
            a[i] = -1;
        }
        for(int i=0; i<a.length; i++) {
            System.out.print(a[i]+" ");
        }
    }
}
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int arr[] = new int[n];
    //To get the values from the user
    for(int i=0; i<n; i++) { // 1 2 3 4 5
        arr[i] = sc.nextInt();
    }
    //Before function call
    for(int i=0; i<n; i++) {
        System.out.print(arr[i]+" ");
    }
    System.out.println();
    UpdateArray(arr);
    //after function call
    for(int i=0; i<n; i++){
        System.out.print(arr[i]+" ");
    }
}
```

Array updation using function: (with return type)

```
import java.util.*;
public class Main {
    public static int[] UpdateArray(int a[]) {
        for(int i=0; i<a.length; i++) {
            a[i] = -1;
        }
        return a;
    }
}
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();
    int arr[] = new int[n];
    //To get the values from the user
    for(int i=0; i<n; i++) { // 1 2 3 4 5
        arr[i] = sc.nextInt();
    }
    //Before function call
    for(int i=0; i<n; i++) {
        System.out.print(arr[i]+" ");
    }
    System.out.println();
    int ans[] = UpdateArray(arr);
    //after function call
    for(int i=0; i<n; i++) {
        System.out.print(ans[i]+" ");
    }
}
```

print odd element in an array using function:

ip:

0	1	2	3	4	5	6	7	8
2	4	8	5	6	7	11	12	13

op:

3	5	7	11	13
---	---	---	----	----

st1: Count No of odd elements

st2: Create a answer array & pass count as a size.

st3: Store the odd values in a answer array.

st4: Return answer array.

```
import java.util.*;
public class Main {
    public static int[] PrintOddElement(int arr[]) {
        int cnt = 0;
        for(int i=0; i<arr.length; i++) {
            if(arr[i]%2!=0) {
                cnt++;
            }
        }
        int ans[] = new int[cnt];
        int idx = 0;
        for(int i=0; i<arr.length; i++) {
            if(arr[i]%2!=0) {
                ans[idx] = arr[i];
                idx++;
            }
        }
        return ans;
    }
}
```

```
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int n = sc.nextInt();//size
    int arr[] = new int[n];
    for(int i=0; i<n; i++) {
        arr[i] = sc.nextInt();
    }
    int res[] = PrintOddElement(arr);
    for(int i=0; i<res.length; i++) {
        System.out.print(res[i]+" ");
    }
}
```

Print minimum & maximum values in an Array

ip:-

2	5	7	1	8	13	12
---	---	---	---	---	----	----

op:-

1	13
---	----

```
public static int[] PrintMinAndMax(int arr[]) {
    int min = arr[0];
    int max = arr[0];
    for(int i=0; i<arr.length; i++){
        if(arr[i]<min){
            min = arr[i];
        }
        if(arr[i]>max){
            max = arr[i];
        }
    }
    int ans[] = new int[2];
    ans[0] = min;
    ans[1] = max;
    return ans;
}
```

```
public static int[] PrintMinAndMax(int arr[]) {
    int min = arr[0];
    int max = arr[0];
    for(int i=0; i<arr.length; i++){
        min = Math.min(min, arr[i]);
        max = Math.max(max, arr[i]);
    }
    int ans[] = new int[2];
    ans[0] = min;
    ans[1] = max;
    return ans;
}
```

Second Largest

Input: arr[] = [12, 35, 1, 10, 34, 1]

Output: 34

Explanation: The largest element of the array is 35 and the second largest element is 34.

```
class Solution {
    public int getSecondLargest(int[] arr) {
        int max1 = arr[0], max2 = -1;
        for(int i=0; i<arr.length; i++){
            max1 = Math.max(max1, arr[i]);
        }
        for(int i=0; i<arr.length; i++){
            if(arr[i]!=max1 && arr[i]>max2){
                max2 = arr[i];
            }
        }
        return max2;
    }
}
```

1295. Find Numbers with Even Number of Digits

Input: nums = [12,345,2,6,7896]

Output: 2

Explanation:

12 contains 2 digits (even number of digits).

345 contains 3 digits (odd number of digits).

2 contains 1 digit (odd number of digits).

6 contains 1 digit (odd number of digits).

7896 contains 4 digits (even number of digits).

Therefore only 12 and 7896 contain an even number of digits.

```
class Solution {
    public int countDigit(int a){
        int cnt = 0;
        while(a!=0){
            cnt++;
            a/=10;
        }
        return cnt;
    }
    public int findNumbers(int[] nums) {
        int cnt = 0;
        for(int i=0;i<nums.length;i++){
            int val = countDigit(nums[i]);
            if(val%2==0){
                cnt++;
            }
        }
        return cnt;
    }
}
```

27. Remove Element

Input: nums = [3,2,2,3], val = 3

Output: 2, nums = [2,2,_,_]

Explanation: Your function should return k = 2, with the first two elements of nums being 2.

It does not matter what you leave beyond the returned k (hence they are underscores).

```
public int removeElement(int[] nums, int val) {
    int idx = 0;
    for (int i = 0; i < nums.length; i++) {
        if (nums[i] != val) {
            nums[idx] = nums[i];
            idx++;
        }
    }
    return idx;
}
```

66. Plus One

Input: digits = [1,2,3]

Output: [1,2,4]

Explanation: The array represents the integer 123. Incrementing by one gives 123 + 1 = 124.

Thus, the result should be [1,2,4].

123 → 123 + 1 = 124

179 → 179 + 1 = 180

199 → 199 + 1 = 200 .

999 → 999 + 1 = 1000 .

```
public int[] plusOne(int[] arr) {  
    int n = arr.length;  
    for(int i=arr.length-1;i>=0;i--){  
        if(arr[i]<9){//199 -> 200  
            arr[i]++;  
            return arr;  
        }else{  
            arr[i] = 0;  
        }  
    }  
    int ans[] = new int[n+1];  
    ans[0] = 1;  
    return ans;  
}
```
